Project ID: **25-26J-511**

Smart Safety Monitoring System For Long Distance Buses In Sri Lanka


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Smart Safety Monitoring System For Long Distance Buses In Sri Lanka

Overall Project Description

This project develops a smart safety system to address the critical issue of accidents involving Sri Lanka's long-distance buses. It integrates real-time monitoring of driver fatigue, route violations, passenger overcrowding, crash detection, and emergency response. Using affordable sensors and software, the system provides immediate alerts to drivers and sends reports to relevant authorities. The goal is to create a safer and more reliable public transport network through practical technology.

Mitigate Overcrowding and Enhance Passenger Safety

Software Engineering

SLIIT FACULTY OF COMPUTING

IT22142696 Dias DGM 25-26J-511

Mitigate Overcrowding and Enhance Passenger Safety

Knowledge Gap & Problem Definition 🧠

There is a critical absence of an

affordable, scalable, and integrated technological solution designed to monitor and mitigate the risks of overcrowding and footboard riding in Sri Lanka's long distance buses in real-time.

→

Existing technologies are not align with unique context of Sri Lanka's fleet, which includes

The chaotic, bidirectional passenger flow on **dual door buses**, where doors are used interchangeably for entry and exit, makes accurate counting extremely difficult **a gap that current monitoring systems do not address**.

Shortcomings of Existing Systems & Our Solution

Shortcoming: Current passenger counting systems rely on expensive video analytics or pressure sensors placed on seats that are not practical for the nation's aging bus fleets.

Our Solution: We use a **cost-effective IoT approach with IR sensors** making it financially suitable for SL's economy

Shortcoming: Solutions in neighboring countries and existing local systems lack central dashboards, violation escalation, and multi-language support, creating a gap between detection and enforcement.

Our Solution: We propose a new, multistakeholder platform includes a mobile app for conductors & passengers , A central dashboard for authorities, enabling data-driven enforcement and proactive safety management.

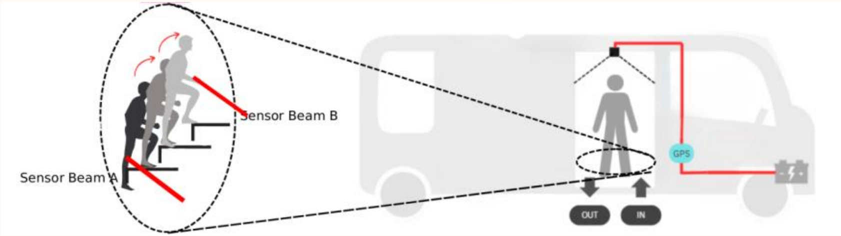
Mitigate Overcrowding and Enhance Passenger Safety

Key Domains of Specialization Utilized 💻

- **IoT & Embedded Systems**: core solution involves designing using an ESP32 microcontroller for edge-based data processing
- **Cloud Computing & Backend Development**: architect a scalable backend using **Firebase Realtime Database** & SQL server will be used for the authority dashboard's data storage
- **Mobile & Web Development**: Requires developing a cross-platform mobile app using **Flutter** for drivers and conductors, alongside a responsive **React.js + node.js** web dashboard for transport authorities

How to Evaluate/Prove Success 📈

- **Accuracy**: The hardware prototype will be designed to achieve ≥95% accuracy in detecting directional passenger flow and footboard occupancy under controlled test conditions.
- **Performance**: The software ecosystem will be evaluated to ensure it provides real-time alerts and occupancy data with a system latency of less than 5 seconds.



Mitigate Overcrowding and Enhance Passenger Safety

Data Availability and Ethical Clearance

Data Source: All primary data is generated in real-time from the on-board IR sensors and GPS modules. No external or pre-existing datasets are required for the core functionality.

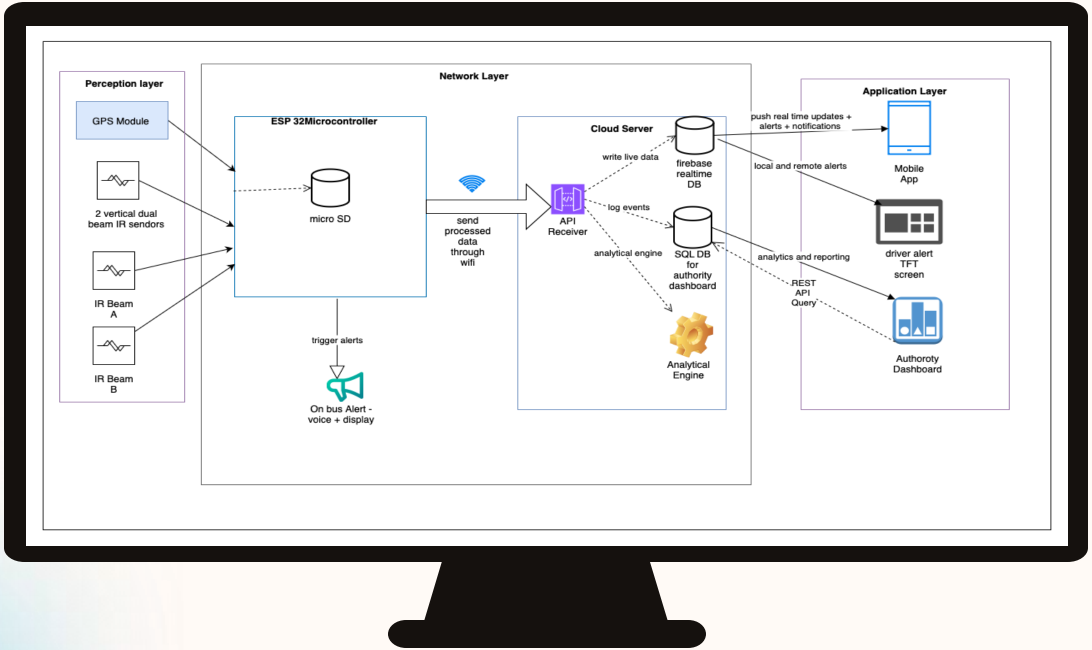
Ethical Clearance: The system is explicitly designed to be privacy-preserving. It aggregates passenger data for statistical summaries only and
does not store any personally identifiable information (PII). This approach avoids privacy concerns associated with camera-based surveillance.

Mitigate Overcrowding and Enhance Passenger Safety

System Overview &

How It Works in a Real-World Scenario

- **Live Monitoring**: Conductor views the live passenger count on the mobile app Passenger view count before onboard.
- **Proactive Alerts**: The app provides early warning as the bus approaches its capacity limit
- **Hazard Detection**: IR sensors automatically detect dangerous footboard riding.
- **Take Action By:** An immediate audio & visual alert warns
- **Automated Enforcement**: The violation is automatically logged with GPS coordinates and a timestamp on the authority dashboard, creating a record to review & enforcement.



Route and Road Violation Detection System

Software Engineering

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Route and Road Violation Detection System

Creative Solution

Current public transport system lack a proper mechanism to detect route and road violations throughout a bus's entire journey.

Our solution is integrated safety system that detects route deviations and road rule violations in real time, providing immediate audible alerts to the driver and automated notifications to authorities, enabling instant correction and effective enforcement.

Knowledge Gap

Integration Gap :No System combines GPS tracking + visual analysis to detect violations like wrong overtaking or route deviation log with locations.

Localization Gap :Foreign trained data models fails in Sri Lanka, we need a local trained model.

Shortcomings of Existing Systems

Isolated System : GPS trackers can't see violations, foreign AI cameras don't understand Sri Lankan roads.

No Real-time Alerts : Systems only records evidence for later review, preventing immediate correction.

Poor Local Accuracy :Global AI models fail on local signs, markings, and driving conditions. forcement and proactive safety management.

Proposed System

Integrated AI + IoT Platforms fuses real time GPS tracking with locally trained AI camera for complete contextual awareness.

Instant audible alerts for drivers, automated notifications to authorities for repeated violations.

Locally optimized YOLO model fine tuned on a custom Sri Lankan road dataset for high accuracy on local violations.

Route and Road Violation Detection System

Application of Specialization

This project integrates core principles from **Software Engineering, Artificial Intelligence, and Internet of Things (IoT)** to solve a complex real-world problem. It applies theoretical knowledge of system design, machine learning, and sensor data fusion into a practical, deployable solution.

Key Domains of Knowledge Utilized

- **Software Engineering**: Full-stack system design, API integration, and database management.
- **Artificial Intelligence (AI)**: Computer Vision (CV), model training and fine-tuning, and sensor fusion logic.
- **Internet of Things (IoT)**: Edge computing, hardware-software integration, and real-time data processing on embedded systems.
- **Cloud Computing**: Scalable data storage, real-time alerting, and dashboard development.

Specific Latest Technologies

- **AI Model**: YOLOv8 for state-of-the-art, real-time object detection.
- **Edge Hardware**: Raspberry Pi 4 for affordable, powerful onboard processing.
- **Cloud and Development**: Firebase Firestore (real time DB), Google Maps API (Geofencing), and React JS (Dashboard)
- **Data**: Custom-trained model on the Roboflow Sri Lankan Road Violations dataset.

Route and Road Violation Detection System

Evaluation & Validation Methods

- **Lab Test**: Measured AI accuracy using part of the Sri Lankan road dataset.
- **Real-World Test**: Used a tabletop model with a toy bus to prove GPS and camera work together in real-time.
- **Measured**: System speed (alert delay) and accuracy.

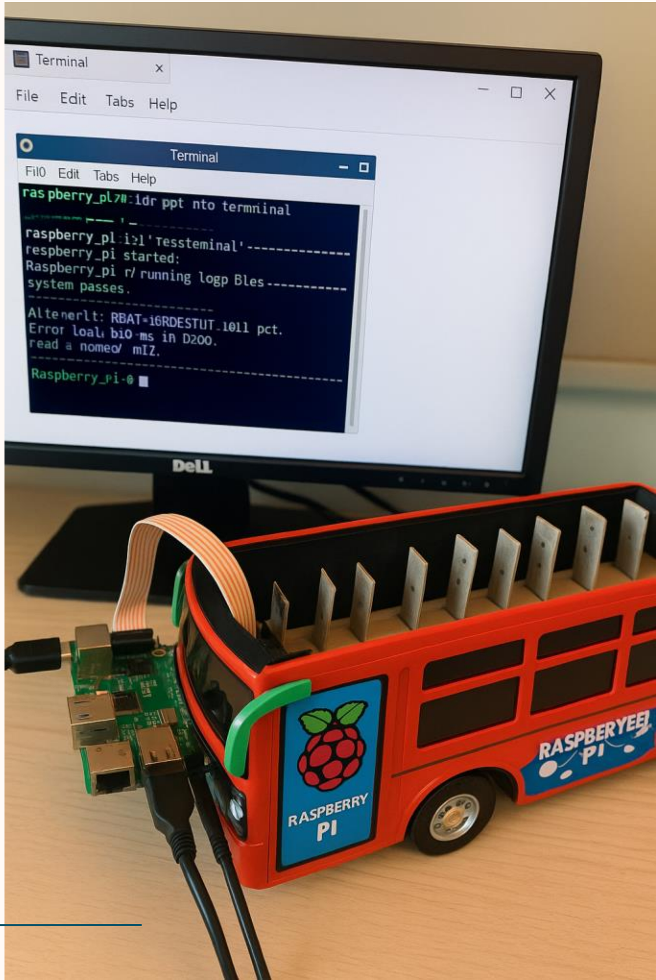
Data Availability

2 Public Sri Lankan datasets from Roboflow for:

- Double line crossing
- Traffic light detection

Ethical Clearance

- Privacy: Video is processed on the device—only violation photos are saved.
- Transparency: Drivers will know about the system.
- Fairness: Using local data makes the AI fair for Sri Lankan roads.



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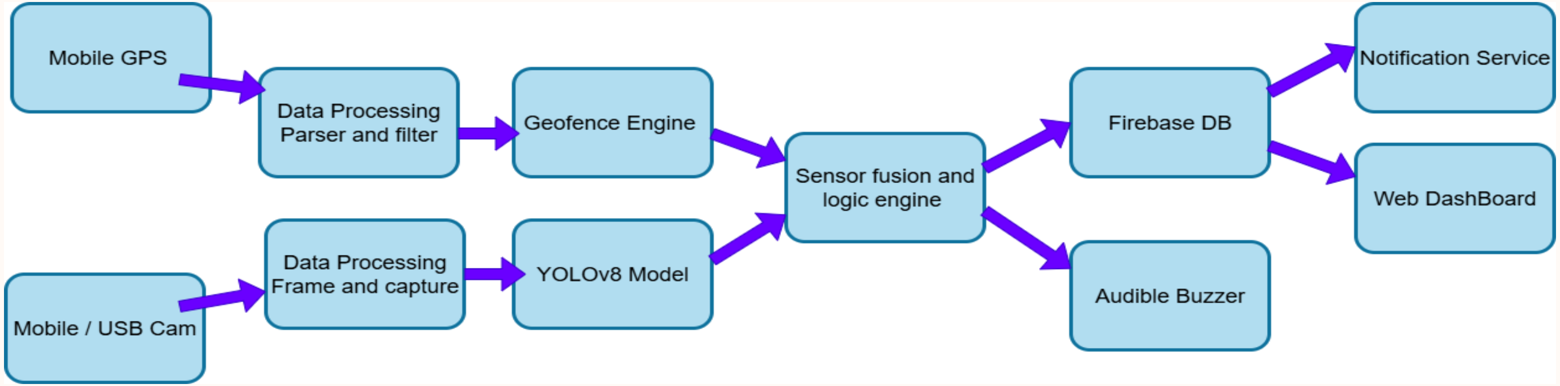
Route and Road Violation Detection System

System Overview



Real World Usage

- **For Drivers**: Immediate audible alerts inside the bus cabin promote safer driving practices.
- **For Fleet Operators:** Web dashboard provides data-driven insights into driver performance and route adherence, enabling better management.
- **For Authorities**: Automated, evidence-based reporting allows for efficient enforcement of traffic laws, targeting repeat offenders.
- **For Sri Lanka**: Contributes to reducing road accidents, improving public trust in transportation, and modernizing the transit system with a scalable, low-cost solution.

Smart, AI-Driven Driver Monitoring for Long-Distance Buses in Sri Lanka

Software Engineering

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Knowledge Gap & Problem Definition 🧠

There is a critical absence of an

affordable, **adaptive**, and **Strong AI Solution** for reliably monitoring and reducing risks of driver fatigue, distraction and no more than 6 continuous driving in Sri Lanka's long-distance bus.

- No automated system to identify drivers violating the law by **driving more than six hours continuously**, which is a major contributor to long distance bus accidents.
- Existing systems are often vision-only, struggling in low-light or occluded conditions common in local buses.
- Fatigue contributes to >30% of annual bus crashes; Manual enforcement detects less than 10% of violations.

Shortcomings of Existing Systems & Our Solution

Shortcoming

- Vision-only approaches fail under poor lighting, occlusion, and environmental noise.
- High cost and complexity of commercial solutions make them unsuitable for local deployment.
- Local systems rarely integrate driving hours, reporting in manually.

Our Solution

- Multimodal monitoring: Fuses video with behavioral and contextual data for robust detection.
- Adaptive AI algorithms: Regular recalibration and retraining tailor detection and alerting to real-world conditions.
- Real-time alerting: Issues immediate audio warnings and notifies operators on authorities.
- Low-cost edge platforms Designed for easy installation on any bus.

Key Domains of Specialization Utilized 💻

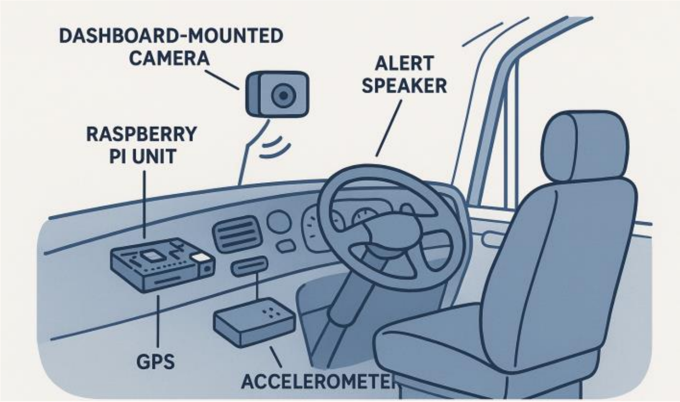
- **Edge AI & Embedded Systems**
  Real-time processing on affordable edge hardware enables in-bus analysis of driver behavior, even without stable internet.

- **Computer Vision & Machine Learning**
  Detection of driver fatigue, distraction and risky using OpenCV and embedded deep learning models. Adaptive learning and threshold calibration for operation across variable lighting and diverse user demographics.

- **IoT & Secure Communication**
  Event-driven architecture automatically logs safety incidents, synchronizes data, and sends alerts over Wi-Fi to operators and central dashboards.

How to Evaluate/Prove Success 📈

- **Accuracy**: System will be design to achieve ≥90% event capture rate in simulated driving scenarios.
- **Performance**: Alerts and logging must occur in ≤2 seconds after risk detection, with full responsiveness on embedded processors under field conditions.



Data Availability and Ethical Clearance

Data Source: Real-time video streams, accelerometer, and GPS data generated within each bus; no reliance on external databases.

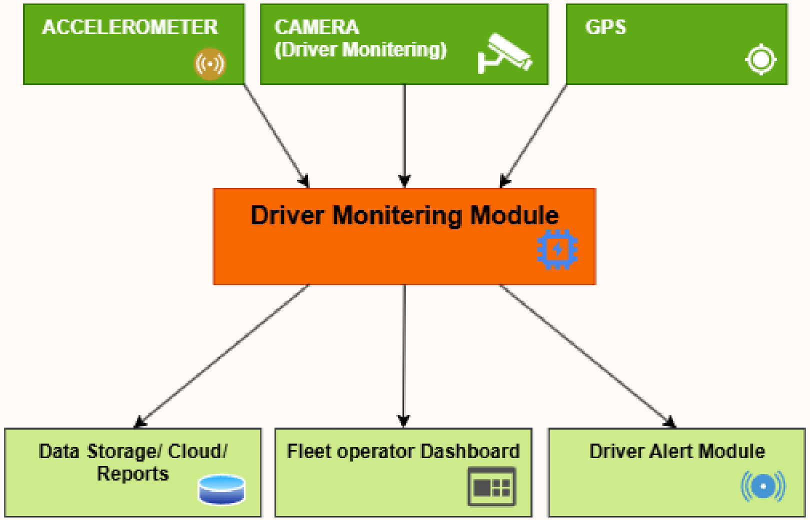
Ethical Design: The system captures only driver facial/upper body data, strictly for safety.

Privacy: Data retention strictly limited (30 days for evidence events); secure logs, access control, and compliance monitoring in place.

System Overview

How It Works in a Real-World Scenario

- **Live Monitoring**: Continuously analyzes the driver's face and head to detect drowsiness, distraction, or dangerous behaviors.
- **Proactive Alerts**: If risky behaviors are detected, the system issues instant to warn the driver and send message to authority
- **Hazard Detection**: Monitors driving patterns via accelerometer ,GPS data to identify aggressive driving.
- **Instant Intervention**: Violations or fatigue are detected, the alert system response less than 2 seconds to help avoid accidents.
- **Automated Enforcement**: Every event is automatically logged with evidence, timestamp, and location. authorities can view incidents.



Emergency Crash Response System for Long-Distance Buses in Sri Lanka

Software Engineering

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Problem & Proposed Solution

Knowledge Gaps & Shortcomings

Eyewitness dependent reporting

- people's memories can be unreliable
- inaccurate and inconsistent
- difficult to get a clear and objective picture

Limited real time location data

- authorities often don't know the exact location
- can delay the emergency response

Weak communication & Coordination

- No proper communication among different agencies
- leads to inefficient coordination

Proposed Solution

Emergency Crash Response System (ECRS)

- Embedded, sensor-driven system
- Automates crash detection in real time
- Classifies crash severity accurately
- Triggers emergency dispatch automatically
- Designed for long-distance buses
- Ensures faster, more reliable post-crash response

Application of Specialization

- **Embedded Systems**
  - Reliable, low-power device
  - can operate independently
  - Always monitors and processes data

- **Artificial Intelligence & Machine Learning**
  - lightweight Convolutional Neural Network (CNN)
  - Avoids false alarms from potholes/braking

- **Sensor Fusion**
  - Kalman Filter – filters out irrelevant or misleading sensor signals

- **Secure Telemetry**
  - IP-based telemetry
  - automated voice calls
  - SMS

Evaluation, Data & Ethical Clearance

Evaluation & Validation Methods

- Lab-Based Prototype Testing
- End-to-End Simulation

Data Availability

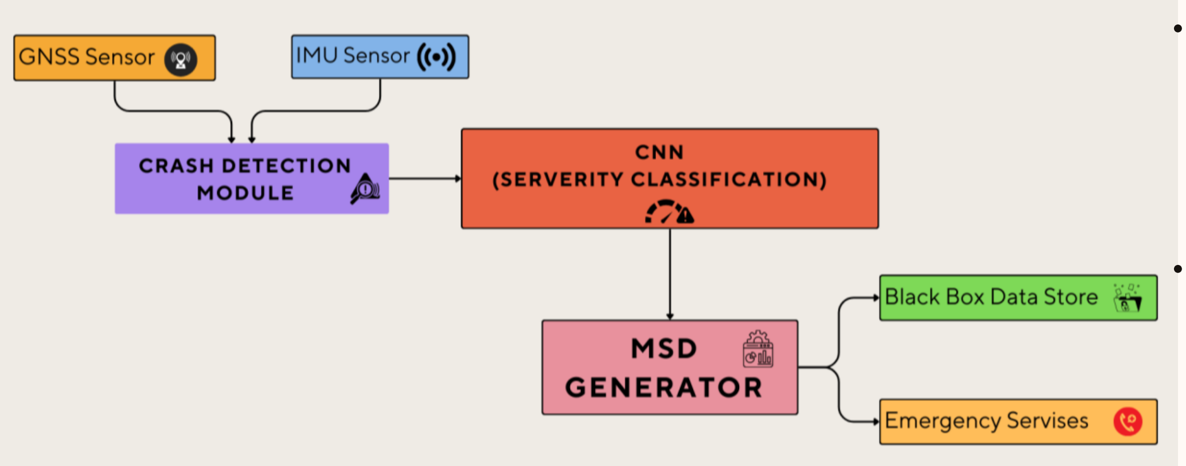
- public IMU datasets
- Vary speed, angles
- Include both severe and minor

Ethical Clearance

- Privacy and security compliant with PDPA
- All sensor and location data encrypted
- Use of Role Based Access (RBAC)

System Overview & Real World Use

System Overview



Real World Usage

- **Auto Alerts**
  - 1990 ambulance
  - 119 police
  - Fleet operators (Transport companies)
- **Severity Based Triage**
  - include crash severity level
  - (minor, moderate, severe)
  - resources are allocated based on urgency
- **Precise GPS Location**
  - Clear accident location
  - Direct navigation

Smart Safety Monitoring System For Long Distance Buses In Sri Lanka

Value Proposition

- **Passengers:** They get to feel safer and travel with peace of mind.
- **Bus Owners:** Helps protect their buses, cut down on extra costs, and build a good name for their service.
- **Authorities:** A simple, low-cost way to keep an eye on buses and enforce rules automatically.

Ability of Commercialization

- The first integrated safety SaaS platform built for Sri Lanka's transport ecosystem.
- Subscriptions + high-margin data monetization (insurance safety scores, urban planning analytics).
- Software-based model easily adapts to new markets (e.g., trucks, school vans) across the Country.

Target Market

- **Main Focus:** Private long-distance bus fleets in Sri Lanka . they'll benefit the most from this system.
- **Next Step:** SLTB buses for pilot projects to test and improve the system.

Investment And Cost Recovery

Investment / Cost

- Labor: R&D Team (Computer Vision, Embedded Systems, Cloud Developers).
- Material: Camera Modules, Sensors (GPS, accelerometer,GNSS), Microcontroller (Raspberry Pi/Arduino), 4G/NB-IoT Module.

Recovery / Pricing

- Hardware Kit Sales: One-time sale of the complete safety monitoring unit.
- SaaS Subscription: Recurring fee for cloud data analytics, real-time alert monitoring, and detailed compliance reporting.
- After-Sales Services: Installation, maintenance, and software update packages.

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Thanks!

Any Questions ?

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